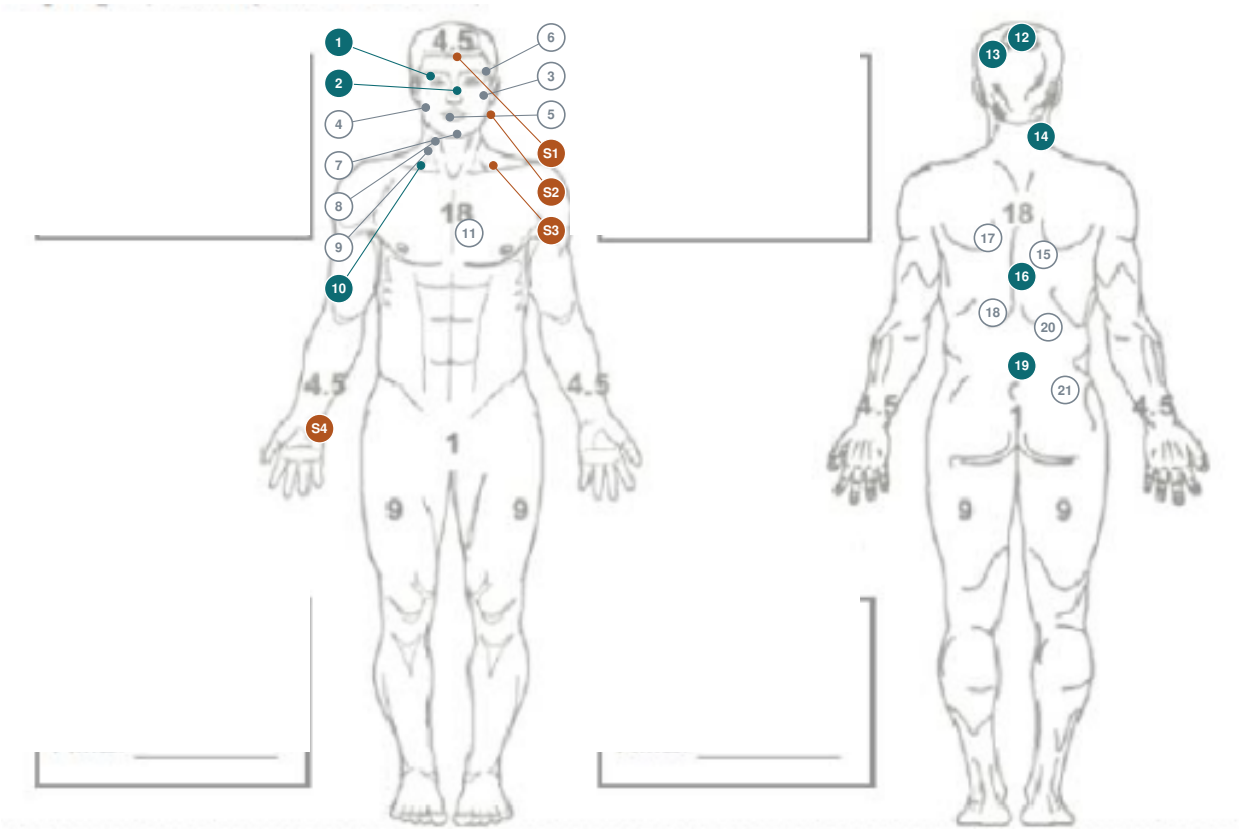
Cerebral circuit anatomy

The 21 components of the Model 6 outflow segmentation, pinned to the body. Filled pins are governed by a streamed parameter (8 of 21); hollow pins are in the circuit and **never estimated** — hollow means unmeasured, not normal.



● governed by a streamed parameter ○ in the circuit, never estimated ● sensor placement site

The pins are schematic. They mark the anatomical region each lumped element stands for, not a surgical landmark: `vertebral_plexus` is one node for a network spanning the whole spinal canal, and `systemic_venous` is a boundary condition rather than a place. The tourniquet fields of the source template are masked — this is an engineering diagram, not a casualty card, and it carries no triage or treatment meaning.

Key

#	Component	State / element	Governed by	Share
1	Cerebral arterial bed	`V_a / C_a`	arterial_compliance	1
2	Intracranial CSF / brain	`Pic (the measured ICP)`	craniospinal_compliance	0.3
3	Cerebral venous compartment	`Pv / V_v`	not estimated	—
4	Arteriolar bed	`Ra`	not estimated	—
5	Capillary bed	`Rc`	not estimated	—
6	Cerebral venous outflow	`Rv`	not estimated	—
7	Craniospinal CSF channel	`Rspine`	not estimated	—
8	Sinus to jugular bulb	`Rss`	not estimated	—
9	Jugular bulb	`P_j`	not estimated	—
10	Internal jugular vein	`Rijvs (tube law)`	jugular_resting_area	1
11	Systemic venous return	`P_sys`	not estimated	—
12	Dural venous sinus	`Psinus`	sagittal_sinus_pressure	1
13	CSF absorption, cranial	`Ro = 0.069*Ro_tot`	csf_outflow_resistance	0.069
14	Vertebral venous collateral	`Rvvs`	collateral_outflow_resistance	1
15	Vertebral venous plexus	`P_vv`	not estimated	—
16	Spinal dural sac	`Psas`	craniospinal_compliance	0.7
17	Azygos return	`Razy`	not estimated	—
18	Paravertebral / azygos	`P_px`	not estimated	—
19	CSF absorption, spinal	`Ro_spine = 0.931*Ro_tot`	csf_outflow_resistance	0.931
20	Vertebral plexus drainage	`Rivp / Revp`	not estimated	—
21	Plexus to systemic shunt	`Rt`	not estimated	—

Sensor sites

#	Placement
S1	ICP monitor (bolt or EVD)
S2	Transducer reference (tragus)
S3	Bilateral jugular compression
S4	Arterial line (radial)

Components read from `scripts/circuit.py` (9 compartments, 12 interfaces, 8 `GOVERNS` entries). Sensor sites follow `SENSORS.md` tier 1. Anatomical placement is not part of the segmentation data.